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ECS 174: Computer Vision
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GitHub Repository: https://github.com/JovinLouie/ECS-174-Project

Road Segmentation

Introduction

Road detection and segmentation from satellite imagery is a longstanding and essential task within the fields of remote sensing, geospatial analysis, and computer vision. Accurate road extraction enables a wide range of real-world applications, including map generation, transportation planning, navigation services, and disaster response. Traditionally, road network identification has relied on manual digitization or semi-automated tools, which can be time-consuming, inconsistent, and unsuitable for large-scale or time-sensitive scenarios.

Recent advances in deep learning, particularly convolutional neural networks (CNNs) and encoder-decoder architectures such as U-Net, have significantly improved the performance of semantic segmentation tasks. These methods allow models to differentiate road pixels from complex background environments that include vegetation, buildings, water bodies, and varying terrain. Automating this process improves scalability, reduces human effort, and increases the reliability of road maps, especially in regions where up-to-date infrastructure data is limited.

This project aims to develop a computer vision model capable of performing road segmentation from satellite imagery. By leveraging a curated dataset and applying state-of-the-art deep learning techniques, the project seeks to demonstrate an efficient, consistent, and practical approach to generating road masks suitable for real-world applications.

Dataset Description

The current project utilizes a satellite imagery dataset downloaded from Kaggle. This dataset was originally created by extracting 400px by 400px images using the Google Maps API, looking at roads in the cities of Los Angeles, California and Boston, Massachusetts. Each sample contains an RGB satellite image paired with a corresponding ground-truth mask that labels road pixels as the positive class and all other terrain as the background. The imagery captures diverse cityscape regions featuring slight variations in weather, lighting, vegetation density, urban development, and road types, all of which provide a realistic setting for training and evaluating a segmentation model.

Dataset Composition

The dataset is composed of 18297 RGB satellite images paired with binary segmentation masks. This culminated in a dataset that is nearly two gigabytes in size. Figures 1 and 2 show a representative satellite image from the dataset alongside its corresponding binary ground-truth mask, where road pixels are labeled in white and all background regions are shown in black.




Figure 1 and 2: boston_0.png (satellite image and ground truth)

Methodology

Data Preprocessing

Due to the large size of our dataset, we had decided not to do any data augmentation. Originally, the dataset was arranged without any training, validation, or testing subsets. Prior to creating our model, we had decided to create a 70-15-15 training/validation/testing split for the original dataset in order to allow for performance monitoring and possible hyperparameter adjustments.

Model Creation

We implemented a standard U-Net encoder–decoder architecture. The encoder consists of four downsampling stages followed by a bottleneck: starting from the 3-channel input, each stage applies two 3x3 convolutions with padding and ReLU activations, then max-pooling. The channel depth increases along the encoder from 64 to 128, 256, 512, and finally 1024 feature maps at the bottleneck. The decoder mirrors this structure with four upsampling stages, each using a 2×2 transposed convolution, followed by concatenation with the corresponding encoder feature maps via skip connections. A final 1x1 convolution maps the 64 decoder channels to a single output channel representing the road probability for each pixel. The resulting network has 31,031,745 trainable parameters. The Unet model was designed to classify every pixel in an image while preserving spatial locations. The encoder extracts general and abstract features while the decoder reconstructs the image using precise locations which are retained by the skip connections. This model's capability for precision and ability to learn a broad range of characteristics makes it an optimal choice for road segmentation.

Training

For training, it was done on a CUDA device taking approximately 4.5 hours to complete. We optimized the network with the Adam optimizer with a learning rate of 0.0005 and a batch size of 8, training for 40 epochs. Because road segmentation is a highly imbalanced problem, we used a composite loss that combines binary cross-entropy and Dice loss. Binary cross-entropy provides stable pixel-wise supervision for classification, while Dice loss measures the overlap between the predicted mask and the ground-truth mask. The BCE element of the loss calculation was given an adjustable weight of 0.5, splitting the responsibility of calculating loss equally between BCE and Dice loss. The BCE term was computed with positive class weights clamped between 1-5 to prevent extreme gradients, and the final loss was computed as an equal-weighted combination of the BCE and Dice components. During training and evaluation, model outputs were passed through a sigmoid and thresholded at 0.5, and the primary metric we tracked was the Dice coefficient on both the training and validation sets, with a final Dice score computed on the test set after training.

Main Results

Across 40 epochs, the model showed steady improvements in its ability to segment road structures from satellite imagery. Training loss consistently decreased while the training accuracy rose throughout the optimization process, indicating that the network continued to learn. In contrast, the test accuracy plateaued around 72–74% suggesting mild overfitting in later epochs. These trends are visualized in Figure 3, which displays the evolution of the training loss, training accuracy, and test accuracy across all epochs. The final model achieved a test Dice coefficient of 72.26%, as shown in Figure 4.

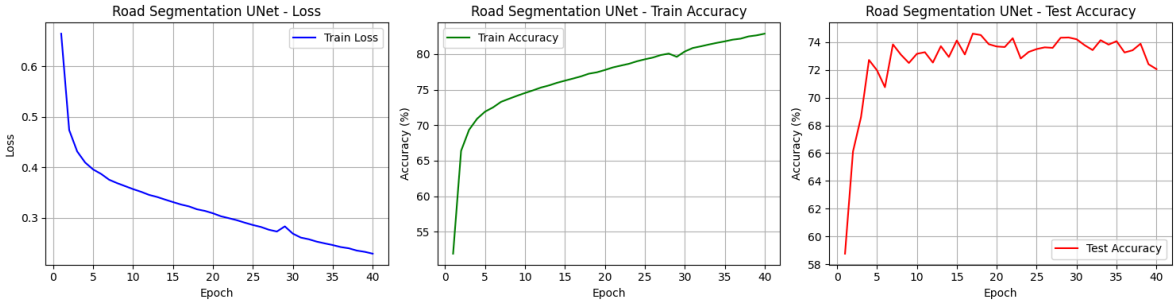


Figure 3: UNet Loss, Training, and Test Accuracy over Epochs

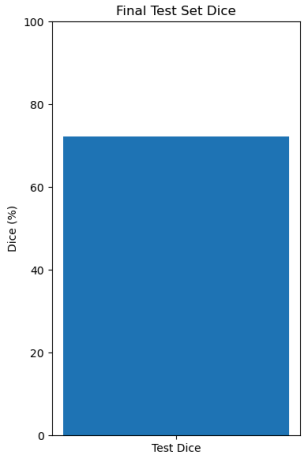


Figure 4: Test Dice for the final 40 epoch model (72.26%)

In addition to the quantitative metrics, we evaluated the model qualitatively by examining predicted masks on individual test samples. Figure 5 and Figure 6 show two representative examples comparing the input satellite image, the ground-truth road mask, and the model's predicted segmentation.

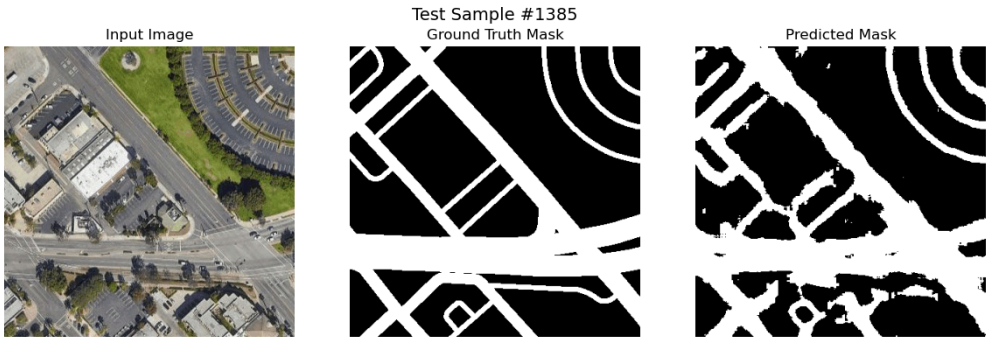


Figure 5: Test Sample #1385

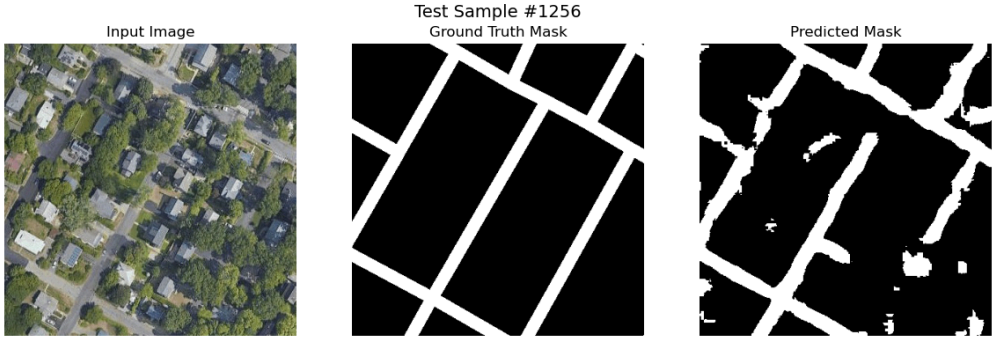


Figure 6: Test Sample #1256

Discussion

The results show that a standard U-Net, trained with a combined BCE + Dice loss, can successfully segment urban road networks from high-resolution satellite imagery with a test Dice score of 72.26%. The model consistently learned meaningful spatial patterns, capturing major road structures and intersections, but struggled with very thin or ambiguous road segments. The gap between training and validation performance suggests mild overfitting. These limitations highlight opportunities for future improvements such as experimenting with U-Net variants, or training on more varied imagery.

Despite these constraints, the model's performance is sufficient for tasks where broad road coverage is more important than pixel-perfect precision. For example, the model could support preliminary map extraction, transportation network modeling, or identifying high-connectivity regions in urban environments. Overall, the project demonstrates that deep learning provides a scalable framework for analyzing high-resolution satellite imagery.

Conclusion

This project implemented and evaluated a U-Net model for road segmentation from satellite imagery, which showed strong performance and produced segmentation masks suitable for practical, high-level geospatial applications. While there is room for improvement through architectural refinements, this project provides a solid foundation for developing more accurate and robust road-segmentation systems. Overall, the project shows that modern computer vision methods can play a valuable role in accelerating map creation, supporting urban analysis, and enabling scalable processing of large satellite imagery datasets in real-world scenarios such as disaster response or infrastructure monitoring.

References

Laborie, T. (2023, July). Satellite Road Segmentation Dataset. Retrieved November 21, 2025 from https://www.kaggle.com/datasets/timothlaborie/roadsegmentation-boston-losangeles/.